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Safety valve, method, and system

Abstract

An insert safety valve, including a housing, a flow tube in the housing, a flapper responsive to positions of the flow tube, an atmospheric chamber in the housing, an element upon the housing, a selectively openable arrangement to prevent or allow pressure communication between the chamber and an environment outside the housing, the arrangement including an opener responsive to movement of the element to selectively open the openable arrangement. A borehole system, including a preexisting downhole structure having a control line, an insert safety valve, in the preexisting downhole structure, the insert safety valve operable with the control line. A method for controlling a wellbore, including running an insert safety valve while preventing fluid communication with an atmospheric chamber of the safety valve, moving the opener, and enabling fluid communication with the atmospheric chamber.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation in part of U.S. application Ser. No. 18/508,802 filed Nov. 14, 2023, the disclosure of which is incorporated by reference herein in its entirety.

BACKGROUND

(1) In the resource recovery and fluid sequestration industries safety valves are often employed for well control. Long service lives are paramount and yet sometimes the life of a safety valve is insufficient for the duty asked of it. In such conditions, insert safety valves are employed to substitute for the function of the previous safety valve. Axiomatically, disposing a replacement valve inside a previous valve reduces available inside diameter, which is generally contrary to desirability. Since insert valves are less expensive than complete workovers, they are not likely to disappear and hence new technologies with enhanced functionality are always desirable.

SUMMARY

(2) An embodiment of an insert safety valve, including a housing, an element disposed upon the housing, a flow tube movably disposed within the housing, a flapper articulated to the housing and responsive to positions of the flow tube relative to the housing, an atmospheric chamber disposed within the housing, a selectively openable arrangement, an opener disposed to selectively open the openable arrangement to allow pressure communication between the atmospheric chamber and an environment outside of the housing, the opener including an opener being responsive to movement of the element on the housing.

(3) An embodiment of a borehole system, including a preexisting downhole structure having a control line, an insert safety valve, disposed within the preexisting downhole structure, the insert safety valve configured to operate based upon input from the control line.

(4) An embodiment of a method for controlling a wellbore, including running the insert safety valve to a location in a wellbore, preventing fluid communication with the atmospheric chamber of the safety valve during running, landing the insert safety valve, moving the opener, and enabling fluid communication with the atmospheric chamber.

(5) An embodiment of a wellbore system, including a borehole in a subsurface formation, a string in the borehole, and an insert safety valve, disposed within or as a part of the string.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The following descriptions should not be considered limiting in any way. With reference to the accompanying drawings, like elements are numbered alike:
- (2) FIG. 1 is a section view of an insert valve as disclosed herein;
- (3) FIG. 2 is a schematic view illustrating the control system of the insert valve of FIG. 1;
- (4) FIG. 3 is a perspective view of a selectively openable arrangement fluidically associated with an atmospheric chamber of the valve of FIG. 1, the openable arrangement in a closed position;
- (5) FIG. 4 is a perspective view of the selectively openable arrangement of FIG. 3, the openable arrangement in an open position;
- (6) FIG. 5 is a perspective view of the openable arrangement 46 and related components;
- (7) FIG. 6 is a sectional view of an alternate openable arrangement in a closed position;
- (8) FIG. 7 is a sectional view of the FIG. 6 embodiment in an open position;
- (9) FIG. 8 is a sectional view of another alternate openable arrangement in a closed position;
- (10) FIG. 9 is a sectional view of the FIG. 8 embodiment in an open position;
- (11) FIG. 10 is a perspective view illustrating a mobilizer for the embodiments of FIGS. 6-9; and
- (12) FIG. 11 is a view of a borehole system including the insert valve as disclosed herein.

DETAILED DESCRIPTION

- (13) A detailed description of one or more embodiments of the disclosed apparatus and method are presented herein by way of exemplification and not limitation with reference to the Figures.
- (14) Referring to FIGS. 1 and 2, a section view of an insert safety valve 10 is illustrated. Insert valve 10 includes a flapper 12, a flow tube 14, a control portion 16 configured to employ control line pressure from a preexisting control line 34 (see FIG. 2) to operate the insert valve 10. The insert valve 10 further includes an atmospheric chamber 22 (see FIG. 2) to reduce actuation pressures required in ways known to the art. The chamber 22, in order to function properly must be maintained free of wellbore fluid during the running process. For this purpose, a selectively openable arrangement 38 that may be configured as a valve 46/56 is disposed in a housing 26 of the insert valve 10 in fluid communication with the atmospheric chamber 22 and between chamber 22 and tubing pressure outside of the housing 26. Atmospheric chamber 22 is also tasked with promoting a failsafe condition of the insert valve by causing the insert valve 10 to be unopenable if closed or to close if open in the event that pressure accesses the atmospheric chamber 22 other than simply by compressing the volume of the chamber 22 due to movement of a flow tube actuating piston 28. Such pressure access is a result of fluid entering the chamber 22 from either a tubing pressure source or a control pressure source.
- (15) The insert valve 10 is run into the borehole to a preexisting downhole structure 30 (see FIG. 2). In most cases, the preexisting downhole structure 30 is configured as a Tubing Retrievable Safety Valve or a Landing Nipple, and it includes a seal bore 32 therein to accept and seal the insert valve 10 therein. The downhole structure 30 will also have a provision for allowing access to the preexisting control line pressure. A preexisting control line 34 was in place to supply control pressure to the preexisting downhole structure 30. Various constructions and methods exist for accessing the original control line pressure and are known to the art.
- (16) Referring to FIGS. 2, 3-5 for specifics and FIG. 1 for location, insert valve 10 includes a first packing arrangement 40, having a tubing pressure side element 42 and a control pressure side element 44. A selectively openable arrangement (38) that may be configured as a valve 46, in an embodiment, is disposed between the element 42 and element 44. The insert valve 10 also may

comprise a second packing arrangement **50** having a tubing pressure side element **52** and a control pressure side element **54**. There also may be a second selectively openable valve **56** disposed between element **52** and element **54**. Openable valves **46** and/or **56** are configured with a valve member **58**, a valve seat **60**, a biaser **62**, and an opener **48** that may be configured as an unseat member **64** in some embodiments. The valve members **58** are initially seated in the valve seats **60** and prevent fluidic contact between the atmospheric chamber **22** and the environment outside of the insert valve **10**. Once the insert valve **10** is landed in the preexisting downhole structure **30** however, the openable valves **46/56** need to be opened to support function of the insert valve **10**. Referring to FIG. 2, the control line **34**, having been accessed by known methods, is available to provide pressure to a volume **68** between the insert valve **10** and the preexisting downhole structure **30**, and bounded by the first and second packing arrangements **40** and **50**, respectively. Pressure applied to the volume **68** is control pressure that is controlled from a remote source by surface personnel or programmable controller. It is this pressure that is used to open the openable valves **46/56** and also to actuate the valve **10**. Accordingly, the valves **46/56** remain closed and protective of the chamber **22** until a decision is taken to open them through the application of pressure in the control line **34**. Pressurizing volume **68** causes one or both of the control side elements **44** and **54** to be moved along the housing **26** toward the unseat member **64**. In FIGS. 3 and 4, only valve **46** is illustrated with surrounding structure. Valve **56** is a mirror image and so it need not be detailed. Focusing on FIGS. 3 and 4, the valve member **58** is seated on seat **60** and thereby allows no fluid communication between the atmospheric chamber **22** (fluidically the chamber **22** is all of the volume behind the valve member **58**) and the environment outside of the housing **26**. Upon the volume **68** being pressurized, element **44** is pushed leftwardly of the figure toward the valve **46**. Comparing FIGS. 3 and 4, it will be appreciated that the valve member **58** and the unseat member **64** have changed position. The element **44** has pushed the unseat member **64** toward the valve member **58**. In the position of FIG. 4, the unseat member **64** physically lifts the valve member **58** off the seat **60**. This can be seen in FIG. 4. Referring to FIG. 5, the unseat member **64** can be seen to exhibit a wedge **66** that engages a recess **67** of the valve member **58**. The urging of the unseat member **64** closer to the valve member **58** causes the valve member **58** to climb the wedge **67** and come off the seat **60**. Once off the seat, the chamber **22** is fluidly communicated to the space between element **42** and **44** bounded by the seal bore **32**. In this condition, the insert valve **10** may be operated via pressure from the control line. However, if packing **40** or **50** leaks in either direction (from the control side pressure or from the tubing pressure side) fluid will infiltrate the chamber **22** and either cause the valve **10** to close or to prevent the opening of valve **10**. In either case, a leak past packing **40** or packing **50** in either direction (i.e., from the tubing pressure side or from the control pressure side) will allow fluid to move into the chamber **22** and thereby cause a fail-safe condition. The pathways implicated may be appreciated in schematic form in FIG. 2, which includes the components discussed above and also illustrates the flow tube piston **28** that is to be actuated by the control line pressure in volume **68** to force the flow tube **14** to move into contact with the flapper **12** (both illustrated schematically in this figure). The piston **28** is fitted with an upper, middle, and lower dynamic seal **74**, **76**, & **78** respectively. The upper dynamic seal **74** is configured to seal against control line pressure on one side and atmospheric pressure on the opposite side. The middle dynamic seal **76** and lower dynamic seal **78** are equivalent in size and may be configured as rod seals with wellbore pressure acting on one side of each seal and atmospheric pressure acting on the opposite side of each seal. With the middle and lower dynamic seals **76** & **78** being equally sized and located on opposite sides of the piston **28**, a pressure balanced condition is achieved with wellbore pressure acting in equal and opposite directions about the piston. This balanced condition makes the insert valve **10** insensitive to wellbore pressure and thus enables actuation at much lower hydraulic pressures in comparison to conventional subsurface safety valves. As with packing **40** and **50**, a leak across dynamic seals **74**, **76**, and **78** will allow fluid to move into chamber **22** and thereby cause a failsafe condition. The entirety of the chamber

22 is further illustrated in FIG. **2** to establish the boundary of where atmospheric pressure shall reside to enable function of the insert valve **10**.

(17) In an embodiment hereof, the control portion **16** of the valve **10** requires more radial space in the housing **26** than a concentric bore for the flow tube **14** would support. Accordingly, insert valve **10** also includes an offset to the bore for the flow tube **14**. In an embodiment, the offset is about 0.250 inch radially. This is illustrated in FIG. **1**, where the deviation in the inside diameter of the valve **10** can be seen at deviation **70** and back at deviation **72**. This provides the additional space needed in the now eccentric annular space of the housing **26** on one side of the housing.

(18) Referring to FIGS. **6-10**, two alternate embodiments for openable arrangements **38** are illustrated. Either of these alternate embodiments that are configured as penetrator configurations **80** or **82**, may be substituted for openable valves **46** and/or **56**. A housing **84** is similar to housing **26** in that a chamber **22** is retained (with the same function) but the housing **84** rather than having a valve seat **60** (as seen in FIG. **3**), instead is closed in this location either by a rupture disk **86** or by a portion **88** of the housing that remains intact inboard of a recess **90** in the housing **84**, which recess **90** is configured to accept a penetrator assembly **92**. Portion **88** segregates recess **90** from chamber **22** until actuation of the penetrator assembly **92** is effected. Elements **94** and **96** are disposed upon the housing **84** in positions similar to those of the embodiments of FIGS. **2, 3, 4**, and **5**. Element **96** is similar to control side element **44** and functions in the same way with control line pressure causing the element **96** to move along the housing **84**. Movement of element **96** causes a variant opener **48** to move. In the penetrator embodiments, the opener **48** is termed a mobilizer **98**. Mobilizer **98** is an analogue of unseat member **64** that functions similarly to unseat member **64** (i.e. moves responsive to element movement) but by causing movement in the penetrator configurations in an opposite direction (radially inwardly of the housing **84**). Specifically, the mobilizer causes movement of a penetrator **100** of the penetrator assembly **92** radially inwardly. Accordingly, movement of the mobilizer **98** will cause the penetrator **100** to load and then pierce the disk **86** or portion **88**. Piercing the disk **86** or portion **88** will fluidically connect chamber **22** with a fluid source radially outwardly positioned of the housing **84**.

(19) Referring specifically to FIGS. **6** and **7** (FIG. **6** in a preopened condition and FIG. **7** in an opened condition), penetrator configuration **80** comprises a penetrator assembly **92**. Assembly **92** includes a disk cartridge **102** that supports the rupture disk **86** and also supports a bushing **104** for the penetrator **100**. Disk cartridge **102** is sealed to housing **84** upon installation. Bushing **104** helps to ensure the penetrator **100** stays in an aligned position. It will be appreciated that the bushing **104** is illustrated with channels **106**, and the penetrator (**100**) is illustrated with a center passage **108**. These features are both for allowing fluid communication between chamber **22** and fluid radially outwardly disposed of housing **84**, once the disk **86** is penetrated. In various embodiments, one or the other of these features may be used alone, while in other embodiments both may be used together.

(20) It will be appreciated when comparing FIGS. **6** and **7** that element **96** has moved leftwardly of the Figure in FIG. **7**. The cause of this movement is identical to that for above described embodiments. The mobilizer **98** is also moved leftwardly due to contact with the element **96**. Referring to FIG. **10**, it can be understood what occurs when the mobilizer **98** moves leftwardly of the figure. Specifically, mobilizer **98** includes an angled surface **110** that interacts with a shoulder **112** on the penetrator **100**. As the mobilizer **98** moves closer to the penetrator **100**, the shoulder **112** climbs the surface **110** thereby urging the penetrator into contact with and into rupture of the rupture disk **86** or portion **88**.

(21) Referring to FIGS. **8** and **9** another penetrator assembly **92** only includes a bushing **114**, the penetrator **100**, and the mobilizer **98**. The difference in the FIG. **6** and FIG. **8** embodiments, is the portion **88** that maintains the seal between chamber **22** and the exterior environment outside of housing **84**. No other seals are needed and hence the disk **86** and disk cartridge **102** are absent. It is noted that the bushing **114** includes the channels **106** for fluid flow when the portion **88** is opened.

Further, it is to be understood that while not illustrated in FIGS. 8 and 9, the penetrator **100** in this embodiment may include the passage **108** (which would appear identically to that shown in FIGS. 6 and 7). The penetrator **100** may be spike shaped or blade shaped as desired to ensure the penetrator effectively cuts through the portion **88**. In other respects the embodiment of FIGS. 8 and 9 works the same way that the embodiment of FIGS. 6 and 7 works.

(22) Referring to FIG. 11, a borehole system **120** is illustrated. The system **120** comprises a borehole **122** in a subsurface formation **124**. A string **126** is disposed within the borehole **122**. An insert safety valve **10** as disclosed herein is disposed within or as a part of the string **126**.

(23) Set forth below are some embodiments of the foregoing disclosure:

(24) Embodiment 1: An insert safety valve, including a housing, a seal element disposed upon the housing, a flow tube movably disposed within the housing, a flapper articulated to the housing and responsive to positions of the flow tube relative to the housing, an atmospheric chamber disposed within the housing, a selectively openable boundary, an opener disposed to selectively open the openable arrangement to allow pressure communication between the atmospheric chamber and an environment outside of the housing, the opener being responsive to movement of the seal element on the housing.

(25) Embodiment 2: The safety valve as in any prior embodiment, further comprising a penetrator.

(26) Embodiment 3: The safety valve as in any prior embodiment, wherein the penetrator assembly defines a passage therein.

(27) Embodiment 4: The safety valve as in any prior embodiment, wherein the selectively openable boundary includes a disk cartridge and a disk.

(28) Embodiment 5: The safety valve as in any prior embodiment, wherein the selectively openable boundary includes a recess and a portion of the housing separating the recess from the atmospheric chamber within the housing.

(29) Embodiment 6: The safety valve as in any prior embodiment, wherein the selectively openable boundary includes a valve member.

(30) Embodiment 7: The safety valve as in any prior embodiment, wherein the valve member includes a biaser.

(31) Embodiment 8: The safety valve as in any prior embodiment, wherein the valve member moves radially inwardly or radially outwardly relative to the housing.

(32) Embodiment 9: The safety valve as in any prior embodiment, wherein the opener is responsive to pressure input.

(33) Embodiment 10: The safety valve as in any prior embodiment, wherein the opener is responsive to a mechanical input.

(34) Embodiment 11: The safety valve as in any prior embodiment, wherein the mechanical input is by a seal element, the seal element being movable based upon application of pressure.

(35) Embodiment 12: The safety valve as in any prior embodiment, wherein the pressure is applied from a control location.

(36) Embodiment 13: The safety valve as in any prior embodiment, wherein the housing further defines a bore for the flow tube that is offset from an axial centerline of the housing.

(37) Embodiment 14: A borehole system, including a preexisting downhole structure having a control line, an insert safety valve as in any prior embodiment, disposed within the preexisting downhole structure, the insert safety valve configured to operate based upon input from the control line.

(38) Embodiment 15: The borehole system as in any prior embodiment, wherein movement of the seal element relative to the housing is by application of pressure from the control line.

(39) Embodiment 16: A method for controlling a wellbore, including running the insert safety valve as in any prior embodiment to a location in a wellbore, preventing fluid communication with the atmospheric chamber of the safety valve during running, landing the insert safety valve, moving the opener, and enabling fluid communication with the atmospheric chamber.

- (40) Embodiment 17: The method as in any prior embodiment, wherein the moving of the opener is by moving the seal element disposed on the housing adjacent the opener.
- (41) Embodiment 18: The method as in any prior embodiment, wherein the moving the opener includes physically moving at least a part of the selectively openable boundary with the opener.
- (42) Embodiment 19: The method as in any prior embodiment, wherein the physically moving includes wedging.
- (43) Embodiment 20: A wellbore system, including a borehole in a subsurface formation, a string in the borehole, and an insert safety valve as in any prior embodiment, disposed within or as a part of the string.
- (44) The use of the terms “a” and “an” and “the” and similar referents in the context of describing the invention (especially in the context of the following claims) are to be construed to cover both the singular and the plural, unless otherwise indicated herein or clearly contradicted by context. Further, it should be noted that the terms “first,” “second,” and the like herein do not denote any order, quantity, or importance, but rather are used to distinguish one element from another. The terms “about”, “substantially” and “generally” are intended to include the degree of error associated with measurement of the particular quantity based upon the equipment available at the time of filing the application. For example, “about” and/or “substantially” and/or “generally” can include a range of $\pm 8\%$ of a given value.
- (45) The teachings of the present disclosure may be used in a variety of well operations. These operations may involve using one or more treatment agents to treat a formation, the fluids resident in a formation, a borehole, and/or equipment in the borehole, such as production tubing. The treatment agents may be in the form of liquids, gases, solids, semi-solids, and mixtures thereof. Illustrative treatment agents include, but are not limited to, fracturing fluids, acids, steam, water, brine, anti-corrosion agents, cement, permeability modifiers, drilling muds, emulsifiers, demulsifiers, tracers, flow improvers etc. Illustrative well operations include, but are not limited to, hydraulic fracturing, stimulation, tracer injection, cleaning, acidizing, steam injection, water flooding, cementing, etc.
- (46) While the invention has been described with reference to an exemplary embodiment or embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the invention. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the invention without departing from the essential scope thereof. Therefore, it is intended that the invention not be limited to the particular embodiment disclosed as the best mode contemplated for carrying out this invention, but that the invention will include all embodiments falling within the scope of the claims. Also, in the drawings and the description, there have been disclosed exemplary embodiments of the invention and, although specific terms may have been employed, they are unless otherwise stated used in a generic and descriptive sense only and not for purposes of limitation, the scope of the invention therefore not being so limited.

Claims

1. An insert safety valve, comprising: a housing; a seal element disposed upon the housing; a flow tube movably disposed within the housing; a flapper articulated to the housing and responsive to positions of the flow tube relative to the housing; an atmospheric chamber disposed within the housing; a selectively openable boundary; an opener disposed to selectively open the selectively openable boundary to allow pressure communication between the atmospheric chamber and an environment outside of the housing, the opener being responsive to movement of the seal element on the housing.
2. The safety valve as claimed in claim 1, further comprising a penetrator.
3. The safety valve as claimed in claim 2, wherein the penetrator defines a passage therein.

4. The safety valve as claimed in claim 1, wherein the selectively openable boundary includes a disk cartridge and a disk.
 5. The safety valve as claimed in claim 1, wherein the selectively openable boundary includes a recess and a portion of the housing separating the recess from the atmospheric chamber within the housing.
 6. The safety valve as claimed in claim 1, wherein the selectively openable boundary includes a valve member.
 7. The safety valve as claimed in claim 6, wherein the valve member includes a biaser.
 8. The safety valve as claimed in claim 6, wherein the valve member moves radially inwardly or radially outwardly relative to the housing.
 9. The safety valve as claimed in claim 1, wherein the opener is responsive to pressure input.
 10. The safety valve as claimed in claim 1, wherein the opener is responsive to a mechanical input.
 11. The safety valve as claimed in claim 10, wherein the mechanical input is by the seal element, the seal element being movable based upon application of pressure.
 12. The safety valve as claimed in claim 11, wherein the pressure is applied from a control location.
 13. The safety valve as claimed in claim 1, wherein the housing further defines a bore for the flow tube that is offset from an axial centerline of the housing.
 14. A borehole system, comprising: a preexisting downhole structure having a control line; an insert safety valve as claimed in claim 1, disposed within the preexisting downhole structure, the insert safety valve configured to operate based upon input from the control line.
 15. The borehole system as claimed in claim 14, wherein movement of the seal element relative to the housing is by application of pressure from the control line.
 16. A method for controlling a wellbore, comprising: running the insert safety valve as claimed in claim 1 to a location in a wellbore; preventing fluid communication with the atmospheric chamber of the safety valve during running; landing the insert safety valve; moving the opener; and enabling fluid communication with the atmospheric chamber.
 17. The method as claimed in claim 16, wherein the moving of the opener is by moving the seal element disposed on the housing adjacent the opener.
 18. The method as claimed in claim 16, wherein the moving the opener includes physically moving at least a part of the selectively openable boundary with the opener.
 19. The method as claimed in claim 18, wherein the physically moving includes wedging.
 20. A wellbore system, comprising: a borehole in a subsurface formation; a string in the borehole; and an insert safety valve as claimed in claim 1, disposed within or as a part of the string.
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